# GUILHERME HENRIQUE RISSON SILVA

Software Engineer Student | Backend & Data | Python | PostgreSQL | FastAPI | LLM
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## SUMMARY

Software Engineering student at Uninter and data analyst at Vitru Educação, where I build the Python systems behind UniCesumar's academic planning. The main one is the faculty allocation optimizer (OR-Tools, FastAPI, React), which assigns about 200 instructors to more than 350 classes per module in minutes. In open-source projects I build backends and LLM applications: Job Hunters (FastAPI, Next.js, PostgreSQL), RAG with hybrid search, and an agent that publishes videos with checked sources.

## EXPERIENCE

**Vitru Educação (UniCesumar)** Hybrid, Maringá, Brazil
Mid-level Data Analyst, Academic Planning 03/2026 – Present

- Built the faculty allocation optimizer (Python, OR-Tools CP-SAT, FastAPI, React, TypeScript), which assigns about 200 instructors to more than 350 classes per module in minutes, work that used to take weeks in spreadsheets.
- Built the 2027 student enrollment forecasting model (Python, DuckDB, Parquet, Power BI), with 50 automated quality checks and error below 2% against actuals.
- Maintain in production the weekly curriculum pipeline for UniCesumar and Uniasselvi (Python, SQL Server), which reads 6 spreadsheet layouts through 4 validation layers.
- Moved the planned-vs-actual audits from Power Query to Python, over datasets of up to 40 million rows, with Power BI dashboards for managers and leadership.
- Rewrote in Python and Marimo the lab class demand vs. tutor capacity model, with a simulator by campus and city whose scenarios cut the shortfall from 27,400 to 21,300 hours.

**Linoforte Móveis** Osvaldo Cruz, Brazil
Assistant Operator 06/2022 – 12/2025

- Set up monitoring of production and quality KPIs, flagging recurring stoppages for corrective action.
- Mapped bottlenecks and downtime and proposed improvements that reduced rework and waste on the line.

## PROJECTS (github.com/guilhermerissondev)

- Job Hunters: job SaaS with FastAPI, Next.js, PostgreSQL/pgvector and LLM. Indexes about 98k jobs across 649 sources and matches with evidence from the CV.
- Teacher Allocation Optimizer: faculty allocation with CP-SAT, FastAPI and React — production system at Vitru, now open source.
- TikTok Viral Generator: agent that tracks news sources, researches, scripts and publishes short pt-BR videos with a source for every fact (Python, Pydantic, SQLite). 4 videos/day, 691 tests.
- Regulatory RAG Eval: question answering over 294 Brazilian Central Bank regulations with source citations (Python, BM25, embeddings, Gemini). Hybrid search raised MRR from 0.361 to 0.644.

## EDUCATION

B.S. in Software Engineering — Uninter — 10/2026 – 2030 (in progress)
AI Software Engineer 4.0 Program (360 h) — Data Science Academy — 09/2026 – 12/2026 (expected)
Postgraduate Certificate in Data Science (360 h) — UniCesumar — 03/2026 – 10/2026
B.S. in Civil Engineering — FAI, Faculdades Adamantinenses Integradas — 2021 – 2025

## SKILLS

Languages: Python, SQL, TypeScript
AI and LLM: RAG, hybrid search (BM25 + embeddings), structured output, LLM evaluation, agents, Gemini API
Backend: FastAPI, Next.js, React, Pydantic, PostgreSQL, pgvector, Docker, pytest, GitHub Actions, Git
Data and ML: pandas, DuckDB, Parquet, SQL Server, scikit-learn, LightGBM, OR-Tools (CP-SAT), Power BI
Spoken languages: professional English, intermediate Spanish